

Zinger Arb-Only Experiment

2-Month Backtest + Live Paper Session

Operator run — 2 Sep 2026

2 September 2026, 18:42 UTC+1

Abstract

This report documents an **arb-only** experiment on Zinger: a **2-month historical backtest** on synthetic Polymarket CLOB books (Jul–Sep 2026 spot data) and a **live paper reset** configured for **forceArbOnly** with loosened gap thresholds. Directional trading is disabled; the bot hunts fee-positive UP+DOWN packages only.

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Executive Summary

Experiment	Result
2mo arb-only backtest (\$10k)	+\$39.82 (+0.40%), 44 arb packages, 100% WR, 0 reloads
Live paper arb-only (reset \$1k)	Running — session <code>session-mtkdu9ic</code> , <code>forceArbOnly: true</code>
Directional trades (both)	0 — arb-only gate active

Caveat: The backtest uses **synthetic CLOB gaps** injected from Binance 1m spot (`bookSim.ts`), not recorded Polymarket order books. Live hit rate will differ. Treat +0.4% over 2 months as a sanity check, not a forecast.

2-Month Historical Backtest (Arb-Only)

2.1 Run Configuration

- Command: `npx tsx scripts/historical-backtest.ts -months=2 -bankroll=10000 -reload=10000 -arb-only -no-fetch`
- Period: ~61 days (2026-07-03 → 2026-09-02)
- Assets: BTC + ETH 1m candles (87,841 bars each, cached)
- Starting bankroll: \$10,000; reload at \$0: \$10,000 (not triggered)
- Engine: `forceArbOnly: true`, governor on, arb-only regime throughout
- Initial arb knobs: `minArbGap` 0.6%, `arbMinMarginPct` 0.3%, `arbBankrollFrac` 18%, `arbMaxUsd` \$100
- Runtime: 63 seconds

2.2 Results

Metric	Value
Final cash	\$10,039.81
Total P&L	+\$39.82 (+0.40%)
Arb packages	44
Directional trades	0
Win rate	100%
Fund reloads	0
Windows scanned	35,135
Scan ticks	105,405
Arb hits (packages locked)	44
Hit rate	~0.042% of scan ticks

Table 1: 2-month arb-only backtest summary

2.3 Bayesian Tuning (In-Run)

After 25 arb packages at 100% WR and +\$21 P&L, the backtest loosened:

- `minArbGap`: 0.006 → 0.0055
- `arbMinMarginPct`: 0.003 → 0.0027
- `arbBankrollFrac`: 0.18 → 0.189

Stratum	Packages	P&L
arb BTC wide fair	20	+\$17
arb ETH wide fair	8	+\$8
arb BTC wide cheap	6	+\$10
arb BTC mid fair	6	+\$3
arb ETH mid fair	4	+\$2

2.4 Top Arb Strata (P&L)

2.5 Comparison to 6-Month Arb Backtest

Run	Period	P&L	Packages
6mo arb-only (1 Sep)	Mar-Sep 2026	+\$60 (+0.60%)	74
2mo arb-only (this run)	Jul-Sep 2026	+\$40 (+0.40%)	44

Table 2: Synthetic backtest comparison — both arb-only, same engine

Saved JSON: data/backtest/backtest-2mo-arb-2026-09-02T17-40-53.json

Live Paper Arb-Only Session

3.1 What Was Done

1. Hard paper reset to **\$1,000**
2. Applied `arbOnlyPaperStrategy()` profile (`modeConfig.ts`)
3. Governor set to **arb-only** regime; DD breaker cleared
4. Bot started; observer attached (`-target=200`, 60s poll)

3.2 Live Configuration

Knob	Value
<code>forceArbOnly</code>	true (no directional entries)
<code>clobArbEnabled</code>	true
<code>minArbGap</code>	0.5%
<code>arbMinMarginPct</code>	0.2%
<code>arbBankrollFrac</code>	12%
<code>arbMaxUsd</code>	\$50
<code>maxArbPackages</code>	6
<code>maxArbPerSlug</code>	3 (stack up to 3 packages per 5m window)
<code>maxOpenPositions</code>	2 (directional cap irrelevant)
<code>governorDrawdownPct</code>	15%
<code>useML / useStrikeForecast</code>	false / false

Table 3: Arb-only paper profile

3.3 Session Identity

- Session ID: `session-mtkdu9ic`
- Started: 2026-09-02 ~17:41 UTC
- Observer log: `data/session-observe-500.jsonl`
- Human log: `logs/session-arb-observe.log`
- Start script: `npm run session:arb`

3.4 How Arb Works (Reminder)

1. When $\text{ask}_{\text{UP}} + \text{ask}_{\text{DOWN}} < 1 - \text{fees} - \text{margin}$, buy equal shares of both legs
2. Walk both ask ladders (not touch-only)
3. Hold to settlement; package net P&L is locked at entry
4. Individual legs show large win/loss at settle — **package net** is what matters

What to Watch (Live Arb Session)

- **Packages per day** — live books are thinner than synthetic; expect far fewer than 44/61d backtest rate unless gaps appear
- **Package-level P&L** — not per-leg settlement display
- **Orphan legs** — if one leg fills and other fails, rollback should unwind (costs both fees)
- **Scan freshness** — stale scan alert means hung loop; restart bot if >90s
- **DD breaker** — at 15% off peak, governor may force arb-only anyway (already arb-only)

Improvement Priorities (Arb Track)

1. **Record real CLOB WS books** — replace synthetic `bookSim` for calibration
2. **Package P&L dashboard card** — aggregate legs, hide confusing per-leg settle losses
3. **Hit-rate telemetry** — ARB SKIP vs PACKAGE LOCKED ratio per hour
4. **Instant CTF merge** on live — lock profit without waiting for window end
5. **Do not re-enable directional** until isolated directional session shows positive expectancy

Files & Commands

Backtest (already run)

```
npx tsx scripts/historical-backtest.ts --months=2 --bankroll=10000 \
  --reload=10000 --arb-only --no-fetch
```

Live arb-only session

```
npm run session:arb
```

Recompile this PDF

```
npm run report:arb-pdf
```

Backtest JSON: data/backtest/backtest-2mo-arb-2026-09-02T17-40-53.json

Backtest log: logs/backtest-2mo-arb.log

This PDF: docs/reports/zinger-arb-run-report.pdf

Status PDF: docs/reports/zinger-status-report.pdf

Synthetic backtest results do not guarantee live performance.
Paper trading only. Not financial advice.